

Robust, Adaptive & Benchmark Attitude Control for a LEO Chaser Rendezvousing with the ISS

Technical Report — ADCS Controller Comparison Tool v1.1.0 (MATLAB / Simulink)

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Abstract. We implement and compare three 3-axis attitude controllers for a chaser spacecraft in a 400 km circular orbit that tracks a line-of-sight reference to the ISS: a boundary-layer quaternion sliding-mode controller (robust, non-adaptive), a parametric adaptive sliding-mode controller that estimates the six inertia parameters online (Slotine–Li regressor with box projection and a freeze-on-saturation anti-windup), and a quaternion PD benchmark. The plant includes rigid-body dynamics with wheel gyroscopic coupling, torque- and momentum-limited reaction wheels, gravity-gradient, aerodynamic, solar-radiation and magnetic disturbance torques, and optional sensor noise. The tool consists of a programmatically generated Simulink model, an App Designer-structured GUI, and a MATLAB reference engine that runs the same library code. In MATLAB R2026a the Simulink harness and the reference engine agree to about 10^{-11}° on every deterministic scenario. Across 7 scenarios $\times$ 3 controllers, both SMC variants reduce the steady-state error roughly $2\times$ relative to PD in docking scenarios. In a persistently exciting scan the adaptive controller is about $75\times$ more accurate than PD. The adaptive controller matches the robust controller when the reference is not persistently exciting. Under persistent excitation, it cuts the tracking error about $8\times$ further. We identify and fix a parameter-windup mechanism during wheel saturation, and characterise the adaptation-gain-versus-excitation trade-off.

Verification status. All results in this report were produced in MATLAB R2026a Update 5 with Simulink 26.1 (2026-09-23). The tables and figures come from the Simulink engine. The MATLAB reference engine reproduces them to at most $2.3\times 10^{-5}^\circ$ on every deterministic scenario, and the sensor-noise scenario differs only through its random streams. The unit-test suite passes 9/9, including the Simulink cross-check. The GUI was not exercised in this validation (notes.md 19). Earlier v1.0.x results from GNU Octave are listed in notes.md §5 for the record.

1. Problem and scope

The chaser performs 3-axis *rotational* control only. Relative translational motion is represented by an analytic closing profile $R(t) = R_f + (R_0 - R_f)e^{-t/T}$ with radial/cross-track offsets that shrink as $(R/R_0)^2$. It defines the docking-axis line of sight but is not integrated as coupled 6DOF dynamics. Four reference modes are available: inertial hold, LVLH hold, docking approach (default), and a 3-axis sinusoidal *attitude scan* about LVLH hold that provides persistent excitation (PE) for the inertia estimator.

2. Models

Quaternions are scalar-first and describe B relative to the orbit-plane inertial frame N, with $C_{BN} = (q_0^2 - q_v^T q_v)I + 2q_v q_v^T - 2q_n[q_v \times]$ [8]. Three orthogonal wheels are aligned with the body axes. The plant is:

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q_dot = 1/2 B(q) w + k_n (1 - |q_raw|^2) q_raw      (constraint-stabilised kinematics)
J w_dot = -w x (J w + h_w) + tau_rw + tau_d         (true inertia J_true)
h_w_dot = -sat(tau_cmd) (zeroed on an axis at |h_w| = h_max), tau_rw = -h_w_dot
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Wheel limits are 0.15 N·m and 15 N·m·s, which corresponds to a derated HR12-25-class wheel [14]. The default inertia is $J = [1200 \ 100 \ -200; 100 \ 2200 \ 300; -200 \ 300 \ 3100]$ kg·m², widely attributed to [6] (attribution unverified; notes.md L1). Disturbances [15][13]: gravity gradient $3n^2 \hat{r} \times (J \hat{r})$; aerodynamic drag at $\rho = 3 \times 10^{-12}$ kg/m³ (USSA-76 gives 2.8×10^{-12} [23]; about 5×10^{-13} to 2×10^{-11} over the solar cycle [26]) with a CP–CG offset; SRP at 4.54×10^{-6} N/m² [25]; and a residual dipole in an aligned-dipole field with $B_0 = 2.97 \times 10^{-5}$ T (IGRF-14 [24]). At a 45° misalignment the model gives $|\tau_{gg}| \approx 3.8 \times 10^{-3}$, $|\tau_{aero}| \approx 1.9 \times 10^{-4}$, $|\tau_{srp}| \approx 8 \times 10^{-6}$ and $|\tau_{mag}| \approx 2 \times 10^{-5}$ N·m.

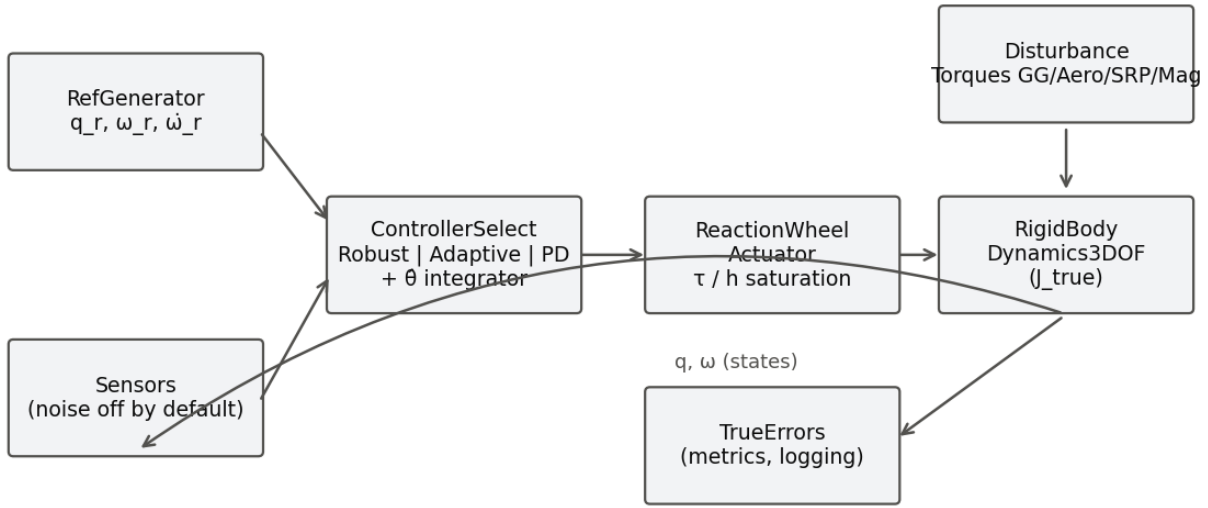


Figure 1. Harness topology. Each block calls the same library function in Simulink (MATLAB Function blocks) and in the reference engine (RK4, same step). Normalisation of q is a separate block to avoid an algebraic loop.

3. Controllers

All controllers use the error quaternion $q_e = q \otimes q_r^*$ (shortest path), the rate error $\omega_e = \omega - C(q_e)\omega_r$, and share a wheel-gyroscopic compensation $+\omega \times h_w$. The two SMC variants share the sliding variable and the Slotine–Li regressor [4][5]:

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s = w_e + Lambda q_ev, w_rv = C(q_e) w_r - Lambda q_ev
Y(w, w_rv, w_rv_dot) theta = J w_rv_dot + w_rv x (J w) theta = [Jxx Jyy Jzz Jxy Jxz Jyz]
(1) Robust SMC : tau = Y theta_nom - K s - eta .* sat(s ./ phi) [1]
(2) Adaptive SMC : tau = Y theta_hat - K s - eta .* sat(s ./ phi)
theta_hat_dot = Proj(-Gamma Y' s), frozen while saturated [4][19]
(3) PD benchmark : tau = -Kp .* q_ev - Kd .* w_e [2][3]
Kp = 2 J_ii wn^2, Kd = 2 zeta wn J_ii

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With the compensation, $J(ds/dt) = (J\omega) \times s + Y(\hat{\theta} - \theta) - Ks - \eta \text{sat}(s/\phi) + \tau_d$. Because $s^T((J\omega) \times s) = 0$, $V = \frac{1}{2} s^T Js + \frac{1}{2} \hat{\theta}^T \Gamma^{-1} \hat{\theta}$ yields the usual bounded-tracking result. Controller 1 is written in regressor form so that Controller 2 differs *only* by $\hat{\theta} \leftarrow \theta_{nom}$; a unit test verifies this to 10^{-12} . Defaults: $\Lambda = 0.05 \text{ s}^{-1}$, $K = \text{diag}(J_{nom})/20$, $\eta = 2 \times 10^{-3} \text{ N}\cdot\text{m}$, $\phi = 10^{-3} \text{ rad/s}$, $\Gamma = 2 \times 10^7$ (5×10^9 in the PE scenario), projection width 60 %, PD with $\omega_n = 0.03 \text{ rad/s}$ and $\zeta = 0.9$.

Anti-windup. In the first verification pass the adaptive controller *lost* to the robust one in both slew scenarios. While the wheels were torque- or momentum-limited for 500–650 s, the closed loop no longer matched the model behind the adaptive law, and $\hat{\theta}$ was integrated onto its projection bounds (78–80 % error). Freezing $\hat{\theta}$ whenever a torque command exceeds τ_{max} or a wheel reaches h_{max} (in the spirit of [19][20]) restored parity; see Figure 5.

4. Implementation

Layer	Content
Simulink	build_ADCS_model.m generates ADCS_ComparisonHarness.slx (ode4, fixed step) and ADCS_Params.sldd. Every MATLAB Function port has a fixed size and type. The diagram is wired from a single table, grouped into seven subsystems with createSubsystem, then every one of the 43 wires is traced across the boundaries, repaired if necessary and re-verified. 19 logged signals appear in logout.
Parameters	Numeric struct parameters SC, RobustGains, AdaptiveGains, BaselineGains and CONTROLLER_SELECT, overridden per run via Simulink.SimulationInput.setVariable (batch of 3 for Compare mode).
Reference engine	simulateADCS_ref.m: RK4 at the same step, running the same library functions, with noise held per step and saturating integrators clamped. It serves as the fallback engine and as the cross-check.
GUI	ADCS_ComparisonApp.m (matlab.apps.AppBase). Tabs: Scenario & ICs, Controller Gains (generated from gainMetadata), Run (single / compare, engine choice), Results (6 plots, metrics table, CSV/figure export). Configs can be saved and loaded.
Tests	9 unit tests (quaternion algebra, reference kinematics, regressor identity, momentum conservation, ablation identity, norm health, metrics, config I/O, Simulink cross-check), a scenario smoke test, and an opt-in wiring-repair check that injects a fault into the model build.

5. Verification

Unit tests (MATLAB R2026a, Simulink 26.1): 9 PASS, 0 FAIL. The Simulink acceptance gate simulates the nominal scenario for 150 s with every controller on both engines. The maximum attitude-error difference is 1.7×10^{-11} , 1.2×10^{-11} and 2.7×10^{-11} (robust, adaptive, PD) against a 0.02° tolerance, and RMS wheel torques agree to 10^{-13} %. Across all runs the largest quaternion-norm deviation is at the 10^{-10} level. Torque-free propagation conserves inertial angular momentum to a relative 10^{-8} over 200 s.

Cross-engine scenario check. All 7 scenarios $\times$ 3 controllers were run on both engines. Each of the 21 pass/fail verdicts is identical on the two engines, and 20 of the 21 runs pass. The residual differences are listed below. They come from the momentum clamp (limited integrator versus post-step clamp) in the saturated slews, and from the independent random streams in the sensor-noise case.

Scenario	max $ \Delta \text{ att_err} $ [deg]	Simulink R/A/PD	Reference R/A/PD
Nominal	1.5e-10	P/P/P	P/P/P
Inertia $\pm 25\%$	2.3e-10	P/P/P	P/P/P
Large error 150°	2.3e-05	P/P/P	P/P/P
Disturbance $4\times$	3.7e-11	P/P/P	P/P/P
Actuator sat.	6.5e-06	P/P/P	P/P/P
Sensor noise	1.3e-02	P/P/P	P/P/P

Scenario	max $ \Delta \text{ att_err} $ [deg]	Simulink R/A/PD	Reference R/A/PD
PE scan $\pm 25\%$	5.8e-10	P/P/F	P/P/F

Table 1. Simulink versus reference engine, MATLAB R2026a. The sensor-noise difference reflects the different noise realisations (Random Number blocks versus randn), not a model difference.

Defects found by the first MATLAB execution and fixed (notes.md I11–I13, D12–D13). (i) With inherited port sizes, Simulink analysed a MATLAB Function block before its input sizes had propagated around the feedback loops, which produced spurious dimension errors; all ports are now fixed-size. (ii) Simulink.BlockDiagram.createSubsystem occasionally re-routed a branched line to the wrong source (for example TrueErr.w_r was fed with q_r). The builder now verifies all wires after grouping and repairs them without leaving orphans. The repair was exercised with an injected fault, and during validation it also caught and fixed a spontaneous occurrence. (iii) Nested Simulink error causes are now reported. A critic review of these changes was accepted after ten findings were addressed.

6. Results

Scenario	Controller	t_s [s]	e_{ss} [deg]	τ_{pk} [N·m]	$\int \dot{\tau} ^2$ [N ² m ² s]	h_{pk} [N·m·s]	τ -sat [s]	h-sat [s]	$\theta^{\wedge} \text{err}$ [%]	Pass
Nominal	Robust SMC	119	0.040	0.150	0.43	2.0	4	0	—	PASS
	Adaptive SMC	119	0.041	0.150	0.43	2.0	4	0	0.5	PASS
	PD Benchmark	110	0.090	0.141	0.32	1.9	0	0	—	PASS
Inertia $\pm 25\%$	Robust SMC	118	0.030	0.150	0.48	2.2	4	0	—	PASS
	Adaptive SMC	118	0.031	0.150	0.48	2.2	4	0	13.8	PASS
	PD Benchmark	110	0.073	0.141	0.36	2.0	0	0	—	PASS
Large error 150°	Robust SMC	591	0.046	0.150	12.47	15.0	492	348	—	PASS
	Adaptive SMC	589	0.046	0.150	12.48	15.0	492	348	1.4	PASS
	PD Benchmark	536	0.085	0.150	10.51	15.0	451	319	—	PASS
Disturbance 4×	Robust SMC	119	0.163	0.150	0.44	2.6	4	0	—	PASS
	Adaptive SMC	118	0.164	0.150	0.44	2.6	4	0	0.5	PASS
	PD Benchmark	111	0.264	0.141	0.33	2.6	0	0	—	PASS
Actuator sat.	Robust SMC	876	0.046	0.075	2.95	7.5	785	657	—	PASS
	Adaptive SMC	876	0.046	0.075	2.95	7.5	785	657	0.3	PASS
	PD Benchmark	833	0.085	0.075	2.63	7.5	747	631	—	PASS
Sensor noise	Robust SMC	119	0.050	0.150	0.44	2.0	4	0	—	PASS
	Adaptive SMC	119	0.051	0.150	0.44	2.0	4	0	0.5	PASS
	PD Benchmark	110	0.101	0.142	0.33	1.9	0	0	—	PASS
PE scan $\pm 25\%$	Robust SMC	132	0.150	0.150	13.97	9.6	6	0	—	PASS
	Adaptive SMC	144	0.019	0.150	13.72	9.6	6	0	1.8	PASS
	PD Benchmark	—	1.461	0.144	18.68	10.4	0	0	—	FAIL

Table 2. Metrics per scenario and controller (Simulink engine, MATLAB R2026a). t_s : time after which the error stays within 0.5°. e_{ss} : mean error over the last 10 % of the run. Pass criteria are scenario-specific (ss-error limit, norm health, finiteness).

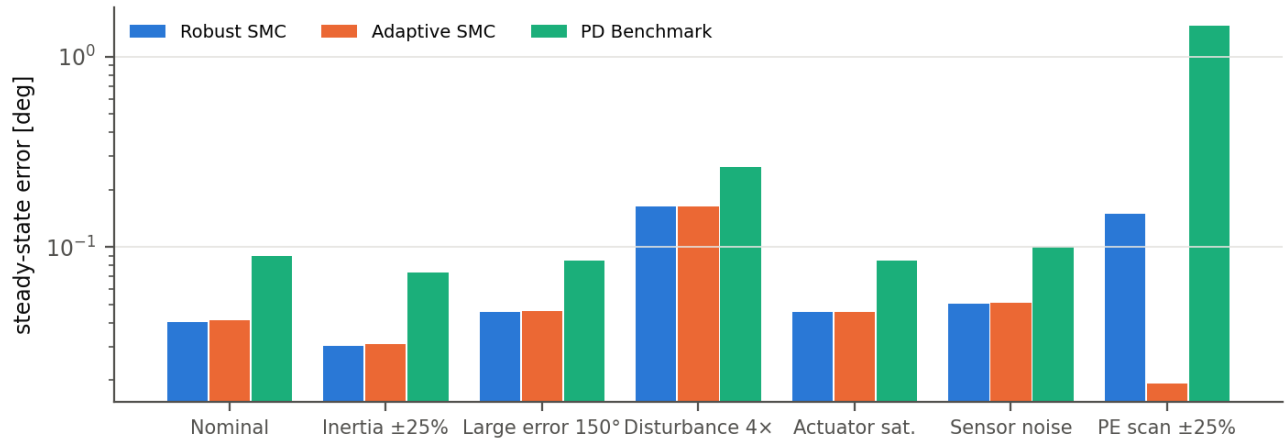


Figure 2. Steady-state error by scenario (log scale).

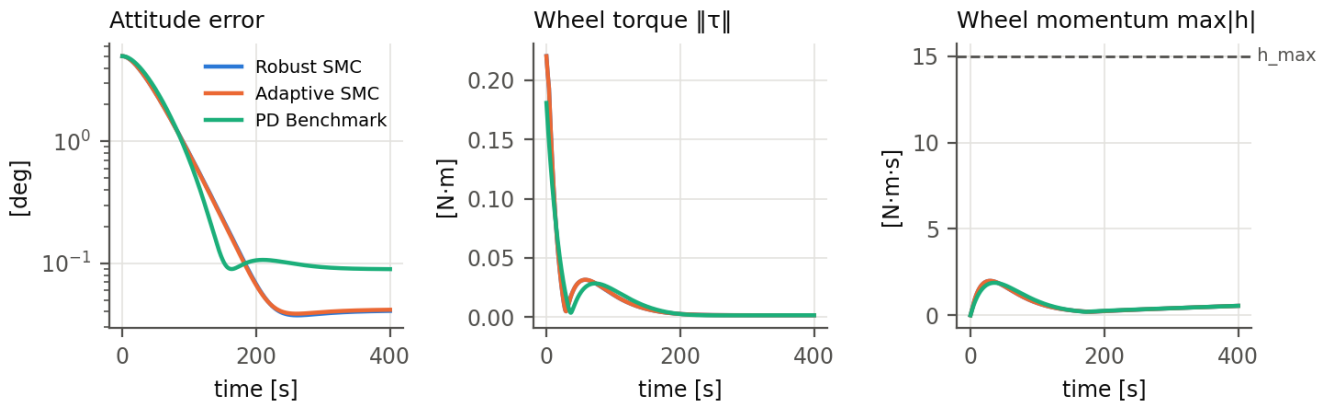


Figure 3. Nominal docking approach (5° initial error). All controllers briefly saturate torque; momentum stays far below $h_{\max}$.

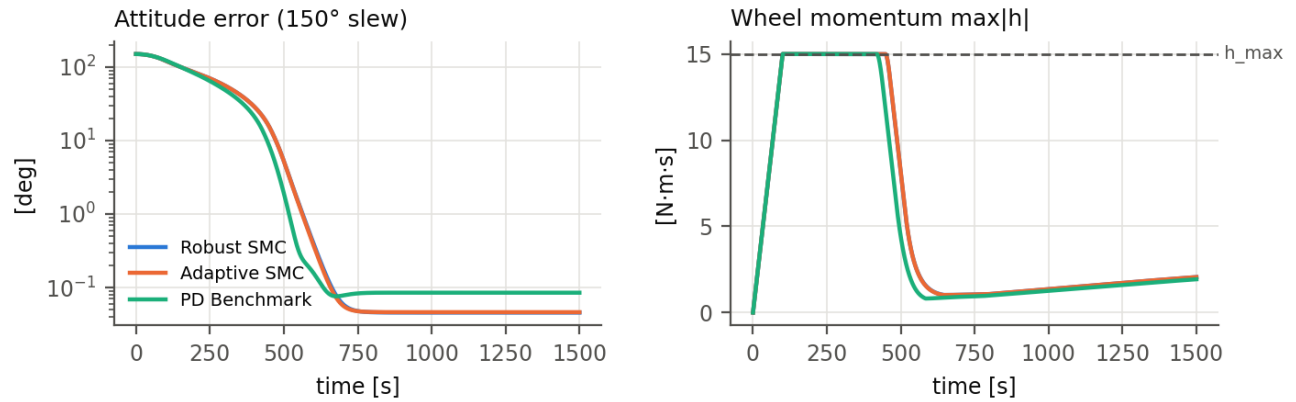


Figure 4. 150° slew. Wheel momentum saturates, so the manoeuvre is rate-limited to about $h_{\max}/J \approx 0.005$ rad/s and settling takes about 550–600 s for every controller.

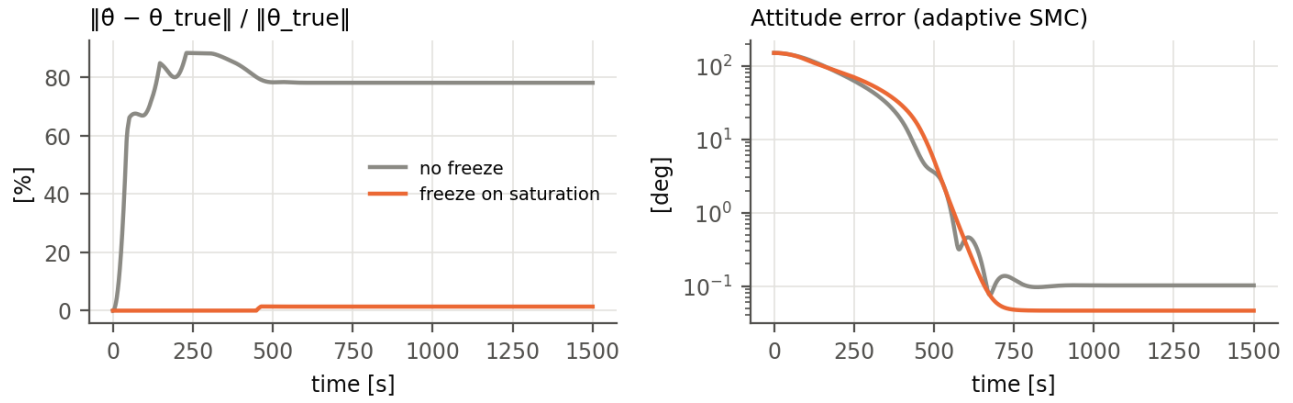


Figure 5. Adaptation windup during the 150° slew. Without the freeze, $\hat{\theta}$ runs to the projection bounds and degrades steady-state tracking. With freeze-on-saturation the estimate is preserved.

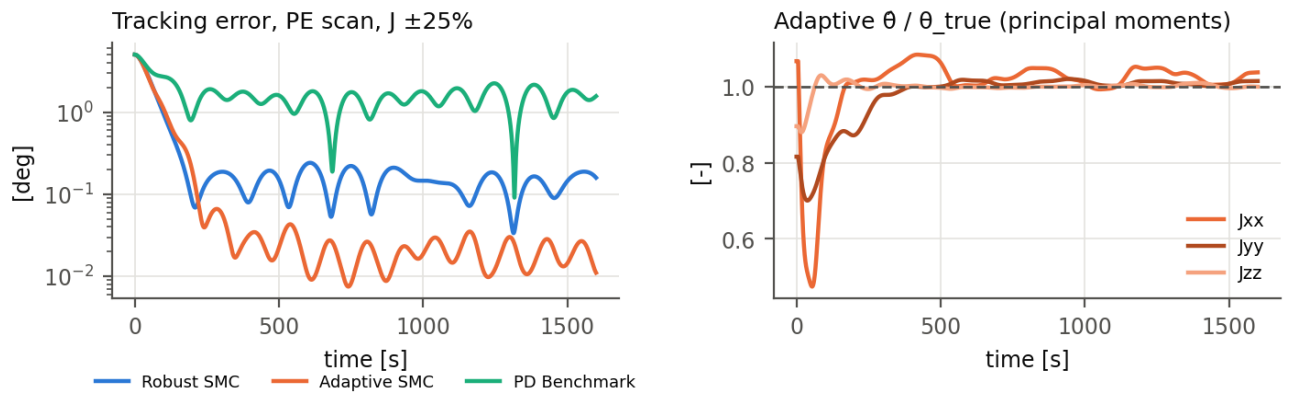


Figure 6. Persistently exciting scan with a $\pm 25\%$ inertia error. The adaptive estimate converges and tracking error falls about $8\times$ below the robust SMC's. PD, lacking feedforward, lags the reference by about 1.5° .

7. Discussion

Robust versus benchmark. Both SMC variants cut the steady-state error relative to PD in every scenario (nominal 0.040° vs 0.090° ; disturbance stress 0.163° vs 0.264°). This comes from the model-based feedforward and the sliding-surface structure; PD's proportional action alone leaves a bias against gravity gradient and drag. PD settles slightly earlier because of its larger proportional authority.

What adaptation buys. In the $\pm 25\%$ docking case, the adaptive and robust controllers are practically identical (0.031° vs 0.030°) and $\hat{\theta}$ hardly moves. The LVLH-referenced reference is nearly stationary, so the regressor is not persistently exciting [9], and the high-gain sliding feedback already absorbs the inertia error. With a PE scan the picture changes: 0.019° (adaptive) vs 0.150° (robust) vs 1.461° (PD), and the principal moments converge to within a few percent.

Adaptation gain versus excitation. A gain sweep ($\Gamma = 10^9$ and 5×10^9 in every non-PE scenario) shows $\hat{\theta}$ drifting 15–49 % from the truth even when $J_{\text{true}} = J_{\text{nom}}$. The estimator absorbs unmodelled disturbance torques, although settling gets faster (81 s vs 119 s). The shipped default is therefore a conservative $\Gamma = 2 \times 10^7$, and the PE scenario raises it explicitly. Excitation-aware adaptation (PE monitoring, composite or concurrent learning, dead zones) is the principled fix and is listed as future work.

Actuator sizing. With $h_{\text{max}} = 15 \text{ N}\cdot\text{m}\cdot\text{s}$ and $J \approx 1200\text{--}3100 \text{ kg}\cdot\text{m}^2$, large slews are momentum-limited whatever the controller, and in practice the actuator dominates the comparison. Performance differences show up in accuracy and robustness, not in slew time.

8. Limitations and future work

Limitations: the model is a rigid body with no momentum management, simplified environment models, an aligned dipole, no eclipse, and ideal sensors by default; the shortest-path sign switch can chatter near 180° [18]; seeded draws (true inertia, noise) are platform- and engine-specific; and the GUI has not yet been exercised in MATLAB. createSubsystem mis-wiring is handled by verification rather than removed at the source. Future work: terminal or prescribed-time SMC [17][22], hybrid hysteretic quaternion feedback [18], input-constrained and excitation-aware adaptation [19][20], a 4-wheel pyramid with faults, coupled 6DOF relative motion [8], MEKF-based estimation [21], and Monte Carlo robustness envelopes. The complete caveat and deviation log, with sources, is in notes.md.

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